doi:10.1038/nature11461

1) Two-box CH_4 and $\delta^{13}C$ model

a) Model set-up

We used a two-box model to investigate the effects of changing sources and sinks on CH_4 and $\delta^{13}C$ in the atmosphere. The model is similar to Tans et al., (1997)[1] and consists of two perfectly mixed boxes representing the two hemispheres. Changes in CH_4 mixing ratios are calculated using the following equations:

$$^{12}CH_4^N(t+\Delta t) = {}^{12}CH_4^N(t) + \left(\sum_i {}^{12}S_i^N(t) + \sum_j {}^{12}k_j\, {}^{12}CH_4^N(t) - \frac{1}{2\tau_{ex}}\, {}^{12}CH_4^N(t) + \frac{1}{2\tau_{ex}}\, {}^{12}CH_4^S(t)\right)\Delta t \quad (1)$$

$$^{12}CH_4^S(t+\Delta t) = {}^{12}CH_4^S(t) + \left(\sum_i {}^{12}S_i^S(t) + \sum_j {}^{12}k_j\, {}^{12}CH_4^S(t) - \frac{1}{2\tau_{ex}}\, {}^{12}CH_4^S(t) + \frac{1}{2\tau_{ex}}\, {}^{12}CH_4^N(t)\right)\Delta t \quad (2)$$

where $^{12}CH_4$ is approximated by CH_4 and $^{12}S_i^N(t)$ and $^{12}S_i^S(t)$ represent the annual source strength of source i in the NH and SH, respectively. $^{12}k_j$ are the first order removal rate coefficients for the three sinks j (OH removal, soil sink, loss to the stratosphere) in units of yr^{-1}. τ_{ex} is the interhemispheric exchange (= 1 yr) time and Δt is the time step, set to 1 year.

The $^{12}CH_4$ emissions, the isotope signatures $\delta^{13}C_i$ of the different source categories i and the kinetic isotope effect KIE_j in the individual sink reactions j are used to calculate the sources ($^{13}S_i$) and removal rate coefficients ($^{13}k_j$) for $^{13}CH_4$:

$$^{13}S_i = {}^{12}S_i * {}^{13}R_{std}\ (\delta^{13}C_i + 1) \quad (3)$$

$$^{13}k_j = \frac{^{12}k_j}{KIE_j} \quad (4)$$

where $^{13}R_{Std}$ = 1.12372% is the $^{13}C/^{12}C$ ratio of the international reference material

Vienna PeeDeeBelemnite (VPDB). These terms are then used to derive the mixing ratio changes of $^{13}CH_4$:

$$^{13}CH_4^N(t+\Delta t) = {}^{13}CH_4^N(t) + \left(\sum_i {}^{13}S_i^N(t) + \sum_j {}^{13}k_j\,{}^{13}CH_4^N(t) - \frac{1}{2\tau_{ex}}\,{}^{13}CH_4^N(t) + \frac{1}{2\tau_{ex}}\,{}^{13}CH_4^S(t)\right)\Delta t \quad (5)$$

$$^{13}CH_4^S(t+\Delta t) = {}^{13}CH_4^S(t) + \left(\sum_i {}^{13}S_i^S(t) + \sum_j {}^{13}k_j\,{}^{13}CH_4^S(t) - \frac{1}{2\tau_{ex}}\,{}^{13}CH_4^S(t) + \frac{1}{2\tau_{ex}}\,{}^{13}CH_4^N(t)\right)\Delta t \quad (6)$$

Finally, the mixing ratios of the individual isotopologues are converted to δ values with:

$$\delta^{13}C = \left(\frac{{}^{13}CH_4/{}^{12}CH_4}{{}^{13}R_{std}} - 1\right) \quad (7)$$

The scenarios for the temporal evolution of the individual CH_4 source categories, their isotopic signatures and the CH_4 sink strengths are used as input to the model (Table S1). The sources are divided in four categories according to their carbon isotopic signatures[2]:

- The pyrogenic sources are the most isotopically enriched ($\delta^{13}C$=-22±3‰) and consist of natural and anthropogenic biomass burning including biofuel burning.
- Biogenic emissions represent the largest sources of pre-industrial CH_4.[3] Boreal and tropical wetlands constitute about half of the biogenic emissions. Other natural sources such as termites, oceans and wild animals are also biogenic. The anthropogenic sources are, in this category, ruminants, rice

paddies, landfills, and CH_4 emissions from wastewater treatment. These different sources are depleted in ^{13}C with isotopic ratios between -55‰ and -65‰. The average value of the biogenic source category varies slightly in the model (between -59.5‰ and -61.5‰) when the relative contributions from the individual sources change.

- The geological sources consist of micro- and macro-seepages, mud volcanoes and geothermal sources[4] with $\delta^{13}C$=-38±7‰.
- The fossil fuel sources include production, consumption and industrial use of oil, coal, and natural gas with an average $\delta^{13}C$=-38±2‰.

OH oxidation in the troposphere is the main CH_4 sink while stratospheric loss via reactions with Cl, O(^1D) and OH radicals and soil removal are in comparison minor sinks. In the model, the effect of the CH_4 sinks on mixing ratio and isotopic composition is characterized by the CH_4 lifetime and the kinetic isotope effect (KIE_j) per sink process j (Table S2).

The model calculates $\delta^{13}C$ and CH_4 time series for each hemisphere at one-year temporal resolution. Measurements[5] and initial modeling studies with the TM3 model[6] show that the inter-polar gradient in CH_4 and its isotopic composition is slightly larger than the observed interhemispheric gradient.

b) Single source scenarios

The single source scenario approach consists of perturbing each type of source individually in order to simulate the impact of single source perturbations on CH_4 and $\delta^{13}C$. To this end, we construct a base scenario based on previous studies[7,8,9] and then perturb the sources one by one to reproduce either the isotope variations (for the centennial scale $\delta^{13}C$ excursions) or mixing ratio variations (for the long-term trend in the mixing ratio). We then investigate the effect of these perturbations on the other quantity (mixing ratio for centennial scale $\delta^{13}C$ excursions and $\delta^{13}C$ for long-term mixing ratio increase)

- Base Scenario

Our model analysis focuses on the period between 100 BC and 1600 AD. We start from a base scenario, in which the relative contributions from the different sources are adjusted to match the average observed mixing ratio ($mr(CH_4)$) and isotope signals during this period ($mr(CH_4)_{NH} \approx 670$ ppb, $mr(CH_4)_{SH} \approx 630$ ppb, $\delta^{13}C_{NH} \approx -47.7‰$, $\delta^{13}C_{SH} \approx -47.2‰$, see Fig. S1). In the base scenario, the sources are maintained constant until 1600 AD. Thereafter, the base scenario follows the hypothesis of former studies[6] assuming a decrease in pyrogenic sources, an exponential increase in biogenic sources (caused by a rise in agricultural emissions) by 1800 AD, followed by an increase of pyrogenic, biogenic and fossil fuel emissions until 2000 AD (Fig. S1).

Note that it is not possible to precisely constrain the relative contribution from the three pre-industrial source categories with only two observables. There are

many possible solutions for the base scenario. Therefore, our base scenario is built on numerous previous studies[6-9]. The strengths of the biogenic and pyrogenic sources are based on [7-9] and are slightly adjusted to fit the $\delta^{13}C$ and CH_4 ice core air datasets from Greenland[10,11] and Antarctica[12,13] (Fig. S1). The source strength of geological sources is maintained constant at 44 Tg yr^{-1} for the entire period. This is consistent with recent estimates for geological sources[4]. For the fossil fuel emissions, the base scenario is derived from the EDGAR 4.1-HYDE inventories. It shows no significant contribution of fossil fuel emissions before 1850 AD, and an exponential increase thereafter. The CH_4 lifetime is held constant at 6.8 years in both hemispheres.

CH_4 sources are not distributed equally between the hemispheres[5], which causes an interhemispheric gradient. In the base scenario, we use the interhemispheric distribution from [9] for the natural emissions and from the EDGAR 4.1-HYDE inventories for the anthropogenic emissions (Table S1). CH_4 mixing ratio and isotopic composition for the SH and NH datasets are not measured in the same institutes, therefore possible intercalibration differences might occur. Given the relatively high scatter of the SH $\delta^{13}C$ data and the lack of data before 1000 AD for these records, we refrain from interpreting the interhemispheric isotope gradient.

- Long-term trends

The CH_4 mixing ratio records reconstructed from ice cores[10-13] show a long-term increase of about 70 ppb between 100 BC and 1600 AD, which corresponds to an increase in annual CH_4 emissions of roughly 28 Tg. Over the same period, we

observe a long-term declining trend in $\delta^{13}C$ of 0.5±0.2‰. To investigate the possible causes of the CH_4 increase, the strengths of individual source categories are changed one by one to fit the CH_4 mixing ratio records. The model then calculates the corresponding long-term trend in $\delta^{13}C$, which can be compared to the measurements (Fig. S2).

The results show that a total increase of 28 Tg of either pyrogenic or geological sources leads to a long-term enrichment in $\delta^{13}C$ between 100 BC and 1600 AD, which does not agree with the observations. In contrast, an increase in biogenic sources agrees reasonably well with both CH_4 mixing ratios and $\delta^{13}C$ (Fig. S2).

A CH_4 trend of the observed magnitude could also be caused by a decrease in the CH_4 lifetime of 1 year (Fig. S2e), but this scenario does not result in a trend in $\delta^{13}C$, and is therefore unlikely to explain the observed long-term decrease. Moreover, the OH abundance in the past cannot reliably be assessed. It depends on several factors (insolation, temperature, humidity and the abundance of other atmospheric compounds), but no long-time temperature trend is observed between 100 BC and 1600 AD. In addition, recent studies[13,14] suggested that the long-term CH_4 increase over the recent millennia is mainly source-driven. In summary, comparison of the $\delta^{13}C$ measurements with the two-box model results strongly suggests that changes in biogenic sources are the most likely explanation for the observed long-term increase in atmospheric CH_4 levels.

-Centennial-scale variability

The three centennial-scale $\delta^{13}C$ excursions must be caused by either a relative increase in the ^{13}C-enriched sources (pyrogenic or geological sources) or/and a decrease from the ^{13}C-depleted sources (biogenic sources). The larger the difference between the isotopic signature of specific sources and the global mean, the higher its "isotope leverage", i.e., the more effective a change in this source is in changing the overall isotopic composition. This is important if we consider the fact that the three centennial-scale anomalies in $\delta^{13}C$ have no clear counterpart (positive or negative) in CH_4 mixing ratios. S3 shows the effects on the CH_4 mixing ratios of changes in the individual sources such that they match the $\delta^{13}C$ observations. Large changes in biogenic or geological sources would be required to explain the isotope excursion as the isotope signature of these sources is relatively close to the mean atmospheric $\delta^{13}C$ value. This would create a strong corresponding signal in the CH_4 mixing ratio scenario, which strongly disagrees with the observations. In contrast, only small changes in pyrogenic emissions are required to explain the observed $\delta^{13}C$ excursions, because CH_4 from these sources is largely enriched in ^{13}C. Therefore, the required additions from pyrogenic emissions are sufficiently small, such that they cannot be resolved in the mixing ratio data (Fig. S3c).

The effect of centennial-scale changes in the CH_4 lifetime has also been investigated. A change in lifetime has an effect on the isotopic composition only during the period of change itself, i.e. when the atmospheric budget is out of equilibrium. If we change the lifetime gradually, it is not possible to reproduce the $\delta^{13}C$ centennial-scale variability (Fig. S4). Moreover, such changes lead to a mismatch with the observed mixing ratios. Hence, changes in lifetime cannot explain

the centennial-scale variability observed in $\delta^{13}C$. We therefore conclude that the centennial scale variability in $\delta^{13}C$ cannot be explained without changes in pyrogenic sources.

A single source scenario approach is adequate to qualitatively identify the most important drivers of the $\delta^{13}C$ variability, but it cannot be used for quantitative source reconstruction as there may be compensating effects from different sources. To overcome this problem, we used a simplified multiple source change scenario, which uses the measured CH_4 and $\delta^{13}C$ to estimate source variations.

c) Multiple source change scenario

In this approach, the measured CH_4 and $\delta^{13}C$ are used to constrain variations in CH_4 sources in each hemisphere. Following Equation 1, the total CH_4 source in the NH can be written as:

$$\sum_i S_i^N(t) = {}^{12}CH_4^N(t+1) - {}^{12}CH_4^N(t) - \sum_j k_j(t)\,{}^{12}CH_4^N(t) + \frac{1}{2\tau_{ex}}\,{}^{12}CH_4^S(t) - \frac{1}{2\tau_{ex}}\,{}^{12}CH_4^N(t)\,\Delta t \quad (8)$$

Similar equations are used for the SH and for $^{13}CH_4$.

Using this approach, a time series of total CH_4 sources can be constructed that leads to a perfect match between model and observations. Similarly, time series of $^{13}CH_4$ emissions are calculated using measured values for CH_4 and $\delta^{13}C$.

- Interpolation of the CH_4 and $\delta^{13}C$ measurements

CH_4 mixing ratio and $\delta^{13}C$ histories at annual resolution are required to derive the annual changes in source strength and isotopic signature (Equation 8). For this purpose, the CH_4 mixing ratio data from WAIS and Law Dome and the $\delta^{13}C$ data from NEEM and EUROCORE for the NH and WAIS and Law Dome for the SH (Fig.1) were smoothed (with a 50 year smoothing filter representative of the age smoothing of air trapped in ice cores, see Fig. S5), and interpolated on a yearly basis.

Since the temporal resolution of the NH CH_4 mixing ratio data is low, the CH_4 mixing ratio data for the NH observations[10,11] is constructed by adding 40 ppb (which is the average difference between the GRIP and Law Dome CH_4 mixing ratio data during this period) to the SH interpolated CH_4 mixing ratio curve. Similarly, 0.5‰ (based on the average different between the NH and SH $\delta^{13}C$ data) was added to the NEEM and EUROCORE $\delta^{13}C$ data in order to obtain the SH $\delta^{13}C$ profile. Only the NH datasets of CH_4 and $\delta^{13}C$ were used to constrain the model.

- Biogenic and pyrogenic emission scenarios

In the top-down approach, as in the single-source approach, CH_4 sources are divided into four categories of which two (fossil fuel and geological) are prescribed in the base scenario. The total source in each hemisphere using Equation 8 can be decomposed into its individual components as:

$$S\,(t) \quad = \quad S_b(t) + S_p(t) + S_g\,(t) + \,S_f(t) \qquad (9)$$

where $S_b(t)$, $S_p(t)$, $S_g(t)$ and $S_f(t)$ represent the strength of emissions from respectively the biogenic, pyrogenic, geological, and fossil sources in each hemisphere.

To derive the total isotopic signature of the CH_4 source per hemisphere we decompose the total source as shown below:

$$\delta^{13}C = \frac{S_b\delta^{13}C_b + S_p\delta^{13}C_p + S_g\delta^{13}C_g + S_f\delta^{13}C_f}{S} \quad (10)$$

of which S_b and S_p are unknown and $\delta^{13}C_b$, $\delta^{13}C_p$, $\delta^{13}C_g$ and $\delta^{13}C_f$ are the isotopic signatures of the biogenic, pyrogenic, geological and fossil fuel source respectively (Table S1). S is the total source strength.

In order to calculate the pyrogenic (S_p) and biogenic (S_b) emission scenarios, we can combine equation 9 and 10 to calculate:

$$S_p = \frac{S(\delta^{13}C_{Tot} - \delta^{13}C_b) - S_g(\delta^{13}C_g - \delta^{13}C_b) - S_f(\delta^{13}C_f - \delta^{13}C_b)}{\delta^{13}C_p - \delta^{13}C_b} \quad (11)$$

$$S_b = S - S_p - S_g - S_f \quad (12)$$

The scenarios calculated using this approach are displayed in Fig.S5.

- Uncertainty analysis

Various sources of uncertainty exist in our CH_4 emission scenarios:

- the observation error (measurement uncertainties and assumed inter-hemispheric gradient)

- the error in the assumed isotope signatures of the different sources and sinks.
- the assumption of holding geological source strength, CH_4 lifetime and inter-hemispheric exchange time constant over time.

A distinction has to be made between "systematic" uncertainties that affect the average strengths of the individual sources, i.e., basically the base scenario as derived above, and "transient" uncertainties, where individual parameters vary with time. Systematic uncertainties are, for example, uncertainties in the average source signatures of the individual source categories or the prescribed strength of the geological source. These uncertainties primarily result in shifting individual source scenarios as a whole up or down, but they are not significant for the magnitude of the transient changes that are investigated in this paper (see below).

"Transient" uncertainties are temporal variations in the model parameters like isotope source signatures, CH_4 lifetime, geological sources or measurement uncertainties. Schaefer and Whiticar, (2008)[20] investigated possible changes in source signatures from different source categories during the last glacial-interglacial transition and found possible changes on the order of 1-2 ‰. We argue that during the climatically-stable late Holocene, isotope variations must have been much smaller than during the glacial transition and should not have exceeded 0.5‰. At this level, transient changes will not significantly alter our source reconstructions. There is no sufficient scientific understanding for estimating changes in the geological source strength, so this parameter is not varied in time. As shown in the single source scenarios above, also transient changes in the lifetime cannot contribute significantly to the centennial scale isotope changes, because they are

effective only during periods where the lifetime changes relatively strongly, but not when a new steady state has been reached. The largest contribution to the transient source reconstructions comes from measurement uncertainties and uncertainties in the assigned interhemispheric differences in $\delta^{13}C$ and CH_4.

To objectively assess the uncertainties in our reconstruction, an ensemble of one thousand simulations was performed, with model input parameters randomly varied within their range of uncertainty. In these simulations we consider both, systematic errors in the geological source strength (44+/-10 Tg yr^{-1}) and isotopic signatures of the sources (Table 1) and systematic as well as transient errors arising from measurement uncertainty by randomly perturbing measurements within their assigned errors (Fig. 1), the inter-hemispheric difference in CH_4 (40+/-10ppb) and the inter-hemispheric difference in $\delta^{13}C$ (0.5+/- 0.2‰). In order to separate the systematic from the transient uncertainties, the calculated emission trends were normalized, so that the remaining variations are only deviations from the average trend. The resulting range of uncertainty for biogenic and pyrogenic sources is reported in Fig. S6.

2) Calculation of the anthropogenic pyrogenic scenario

It is not possible to distinguish natural and anthropogenic pyrogenic activity based on isotope measurements. Therefore, we construct a scenario of anthropogenic pyrogenic activity based on existing estimates of human land use and land use change[15]. We consider anthropogenic CH_4 pyrogenic emissions to be the sum of emissions related to the annual maintenance of crop and pasture land and

the emissions from land clearance[15]. The CH_4 emissions from cropland maintenance ($CH_{4,lm}$) and land clearance ($CH_{4,lc}$) were calculated as shown below:

$$CH_{4,lm} = A \cdot p \cdot C \cdot O \cdot M \cdot N \quad (13)$$

$$CH_{4,lc} = A \cdot H \cdot O \cdot M \cdot N \quad (14)$$

where A is the total area of land under human influence, p the fraction of crop residues and pasture land burnt annually (20%), C the mean value for C in cropland residues per m^2 (200 gC/m^2) [21], O the conversion factor to convert from g C to organic matter (OM) (2.22 gC/gOM), M the conversion factor for converting kg of organic matter combusted to g C emitted as CH_4 (4.8 gC/kgOM), N the conversion factor from g CH_4 to g C (16 g CH_4 / 12g C) and H the estimate of average aboveground C in NH forests (10 kgC/m^2)[21].

The total of the emission from land clearance (conversion to cropland) plus cropland maintenance is defined as the total anthropogenic pyrogenic CH_4 emission shown as a black curve in Fig.S7.

The largest CH_4 flux comes from the annual maintenance burning of crop and pasture land (Fig. S7: blue). A smaller amount of CH_4 emissions is a result of land clearance. (Fig. S7: orange). The maintenance flux shows a rather steady increase over the time period, while additional centennial-scale variability comes from the land clearance flux. Interestingly, these periods appear related to the $\delta^{13}C$ variations. Total emissions range between 6 and 12 Tg yr^{-1} over the time period of interest, which represent roughly 20 to 30% of the total pyrogenic emissions.

3) Area under land use and uncertainty estimates

We estimated total global land use following the methodology used in [22] and [23], where land use is a function of the human population in a country sized region at any given time and the suitability of the land for cultivation. The largest source of uncertainty in the historical land use scenario is the population data that are used to drive the land use model[22]. To provide an estimate of the uncertainty in the land use scenario, we therefore produced additional "lower-bound" and "upper-bound" scenarios (shaded area in Fig. 3) that are based on ranges for past population contained in a global paleodemography database. The paleodemography database is assembled from published estimates of past population for 206 country-sized regions of the world. Methods used to construct this database are detailed in [24]

The "best guess" estimate of total area under human land use has been revised downward about 15% relative to the global totals published in Table 1 of [15] because we 1) improved our datasets of global human population since the original publication, and 2) because we calculate the total land use area on a high-resolution 5 arc-minute grid, instead of the 30' grid used in [15].

Over the period 50 BC to 1,600 AD, our "best guess" land use scenario generally follows the most conservative estimate of human population. Between 1,000 AD and 1,600 AD, however, our best estimate diverges from the lower bound chiefly because of uncertainty in the pre-Columbian population of the Americas. Recent research has demonstrated that the lowest published historical estimates of population in the Americas before 1500 AD, e.g., McEvedy and Jones (1974) [25], are probably large underestimates, and cannot be supported by archaeological and paleoecological evidence for the collapse in populations that occurred after European contact[26].

4) Gas age distribution of NEEM and EUROCORE

The air enclosed in polar ice core does not have a well-defined age but is a mixture of air of different ages, because of mixing within the firn column. It takes several decades before the air becomes totally trapped in the ice, which results in relatively wide gas age distributions for air samples trapped in ice cores[18]. Typical gas age distributions for the NEEM and EUROCORE ice samples were calculated according to [16] and [17] and are shown in Fig. S5. The width of the age spectrum is 32 years for NEEM and 26 years for EUROCORE.

References

1 Tans, P.P., A note on isotopic ratios and the global atmospheric methane budget, *Gl. Biogeochem. Cycles* 11, 77 (1997).

2 Quay, P. et al., The isotopic composition of atmospheric methane, *Global Biogeochem Cy* 13 (2), 445 (1999).

3 Tyler, S. C., Rice, A. L., and Ajie, H. O., Stable isotope ratios in atmospheric CH_4: Implications for seasonal sources and sinks, *Journal of Geophysical Research-Atmospheres* 112 (D3) (2007).

4 Etiope, G., Lassey, K. R., Klusman, R. W., and Boschi, E., Reappraisal of the fossil methane budget and related emission from geologic sources, *Geophys. Res. Lett.* 35 (9) (2008).

5 Miller, J.B. et al., Development of analytical methods and measurements of $^{13}C/^{12}C$ in atmospheric CH_4 from the NOAA/CMDL global air sampling network, *J. Geophys. Res.* 107, 4178 (2002).

6 Monteil, G. et al., Interpreting methane variations in the past two decades using measurements of CH_4 mixing ratio and isotopic composition, *Atmos. Chem. Phys.* 11 (17), 9141 (2011).

7 Houweling, S. et al., Early anthropogenic CH_4 emissions and the variation of CH_4 and $^{13}CH_4$ over the last millennium, *Global Biogeochem Cy* 22 (1) (2008).

8 Ferretti, D. et al., Unexpected changes to the global methane budget over the past 2000 years, *Science* 309, 1714 (2005).

9 Mischler, J. A. et al., Carbon and hydrogen isotopic composition of methane over the last 1000 years, *Global Biogeochem Cy* 23, GB4024 (2009).

10 Chappellaz, Jérôme et al., Changes in atmospheric CH_4 gradient between Greenland and Antarctica during Holocene, *J. Geophys. Res.* 102, NO. D13, 15 (1997).

11 Blunier, T. et al., Variations in atmospheric methane concentration during the Holocene epoch, *Nature* 374 (6517), 46 (1995).

12 MacFarling Meure, C. et al., Law Dome CO_2, CH_4 and N_2O ice core records extended to 2000 years BP *Geophys. Res. Lett.* 33, L14810 (2006).

13 Mitchell, L. E. et al., Multidecadal variability of atmospheric methane, 1000-1800 CE, *J. Geophys. Res.* 116 (2011).

14 Levine, J. G. et al., In search of an ice core signal to differentiate between source-driven and sink-driven changes in atmospheric methane, *J. Geophys. Res.* **116** (2011).

15 Kaplan, J. O. et al., Holocene carbon emissions as a result of anthropogenic land cover change, *Holocene* **21** (5), 775 (2011).

16 Witrant, E. et al., A new multi-gas constrained model of trace gas non-homogeneous transport in firn: evaluation and behavior at eleven polar sites, *Atmos. Chem. Phys. Discussions*, **11**, (23), 029–23 080 (2011).

17 Rommelaere, V., Arnaud, L., and Barnola, J.-M., Reconstructing recent atmospheric trace gas concentrations from polar firn and bubbly ice data by inverse methods, *J. Geophys. Res.*, **102**, 30 069–30 083 (1997).

18 Schwander, J. et al., The Age of the Air in the Firn and the Ice at Summit, Greenland, *J. Geophys. Res.* **98** (D2), 2831 (1993).

20 Schaefer, H., Whiticar, M. J., Potential glacial-interglacial changes in stable carbon isotope ratios of methane sources and sink fractionation, *Glob., biogeochem. cy.*, **22**, GB1001 (2008).

21 Kaplan, J. O., Krumhardt, K. M. and Zimmermann, N. E., The effects of human land use and climate change over the past 500 years on the carbon cycle of Europe, *Glob. Change Biol.*, **18**, 902-914 (2012).

22 Kaplan, J. O., K. M. Krumhardt, E. C. Ellis, W. F. Ruddiman, C. Lemmen, and K. K. Goldewijk (2011), Holocene carbon emissions as a result of anthropogenic land cover change, *The Holocene*, *21*(5), 775-791.

23 Kaplan, J. O., K. M. Krumhardt, and N. Zimmermann (2009), The prehistoric and preindustrial deforestation of Europe, *Quaternary Sci. Rev.*, *28*(27-28), 3016-3034.

24 Krumhardt, K. M. (2010), ARVE Technical Report #3: Methodology for worldwide population estimates: 1000 BC to 1850, ARVE Research Group, École Polytechnique Fédérale de Lausanne,

25 McEvedy, C., and R. Jones (1978), *Atlas of World Population History*, 368 pp., Penguin, London.

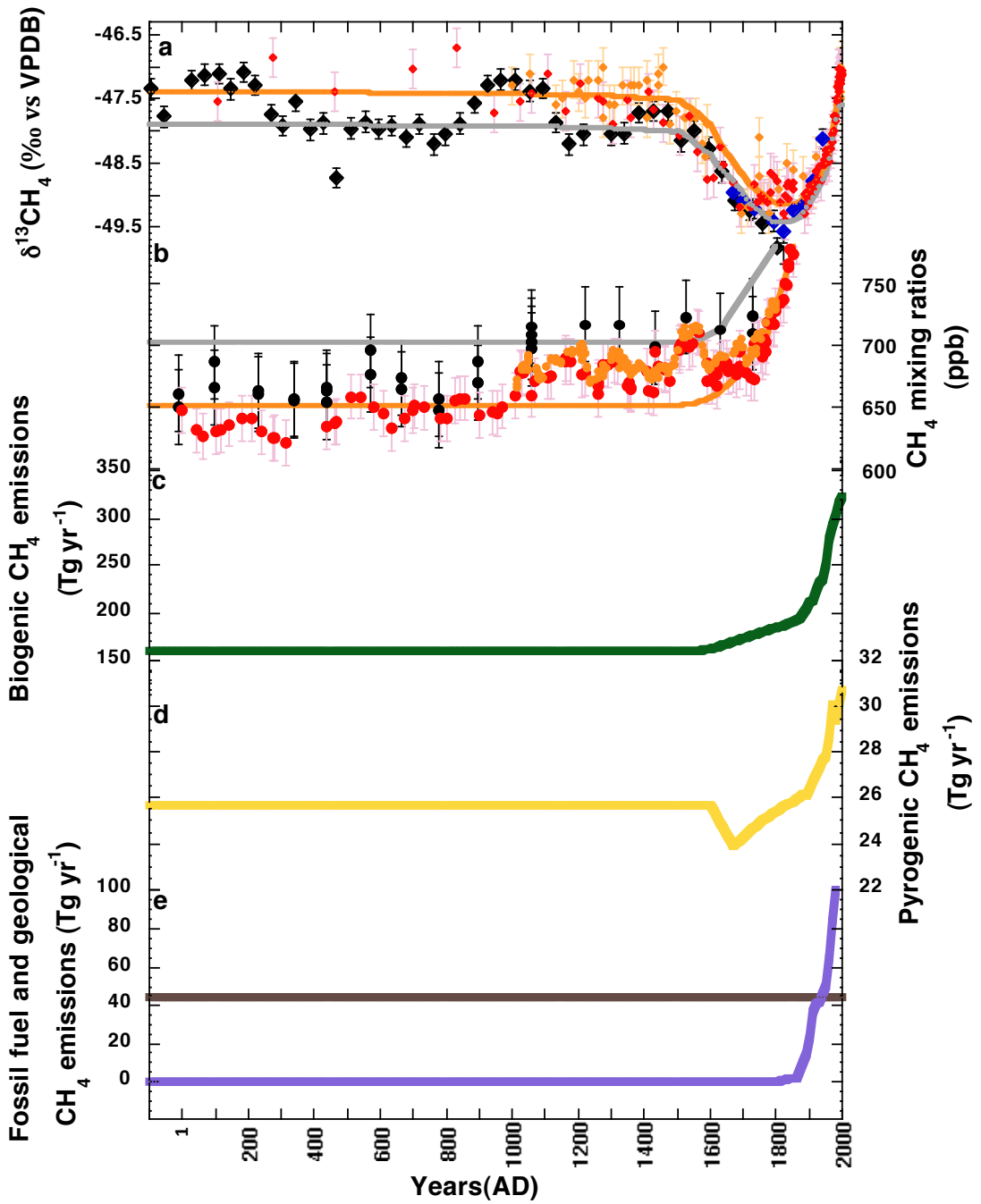
26 Nevle, R.J. and Bird, D.K., 2008. Effects of syn-pandenic fire reduction and reforestation in the tropical Americas on atmospheric CO_2 during European conquest. *Palaeogeography, Palaeoclimatology, Palaeoecology*, *264* (1-2), 25-

Table S1: Range of parameters varied in the model, i.e., CH_4 source strength from geological sources, isotope signatures of the main CH_4 sources and their interhemispheric distribution[7].

Sources	Source strength ($Tg.yr^{-1}$)	$\delta^{13}C$ (‰ vs VPDB)	NH (%)	SH (%)
Pyrogenic		-19 to -25	44±3	56±3
Biogenic		-55 to -65	80±5	20±5
Geological	34 to 54	-31 to -45	90±5	10±5

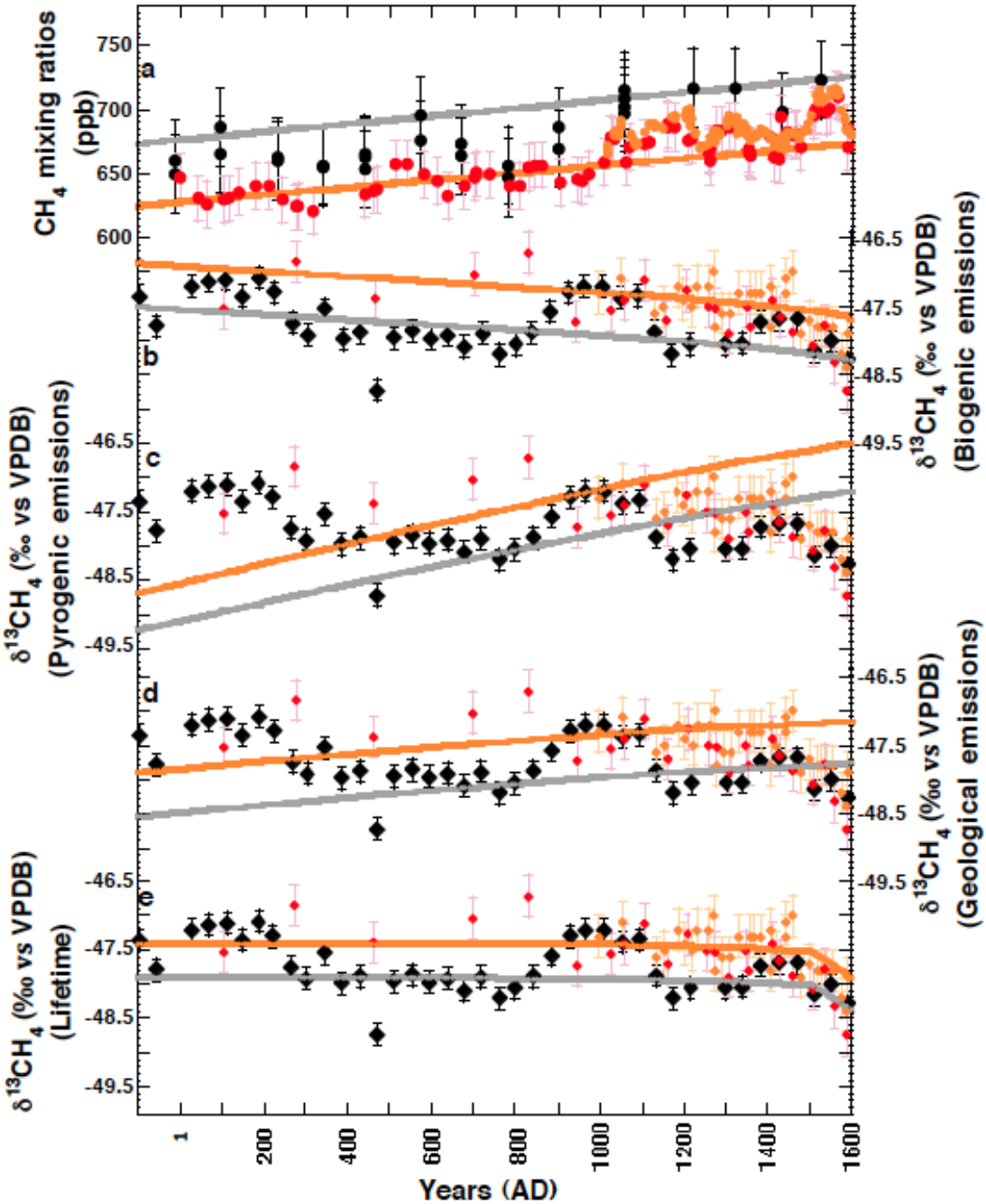
Table S2: Inverse lifetime $1/\tau$ and kinetic isotope effect KIE^{13C} for the three different sinks of CH_4 [9].

SINKS	$1/\tau$	KIE^{13C}
OH oxidation	0.133	1.0054
Soil removal	0.008	1.0180
Statospheric loss	0.006	1.0120



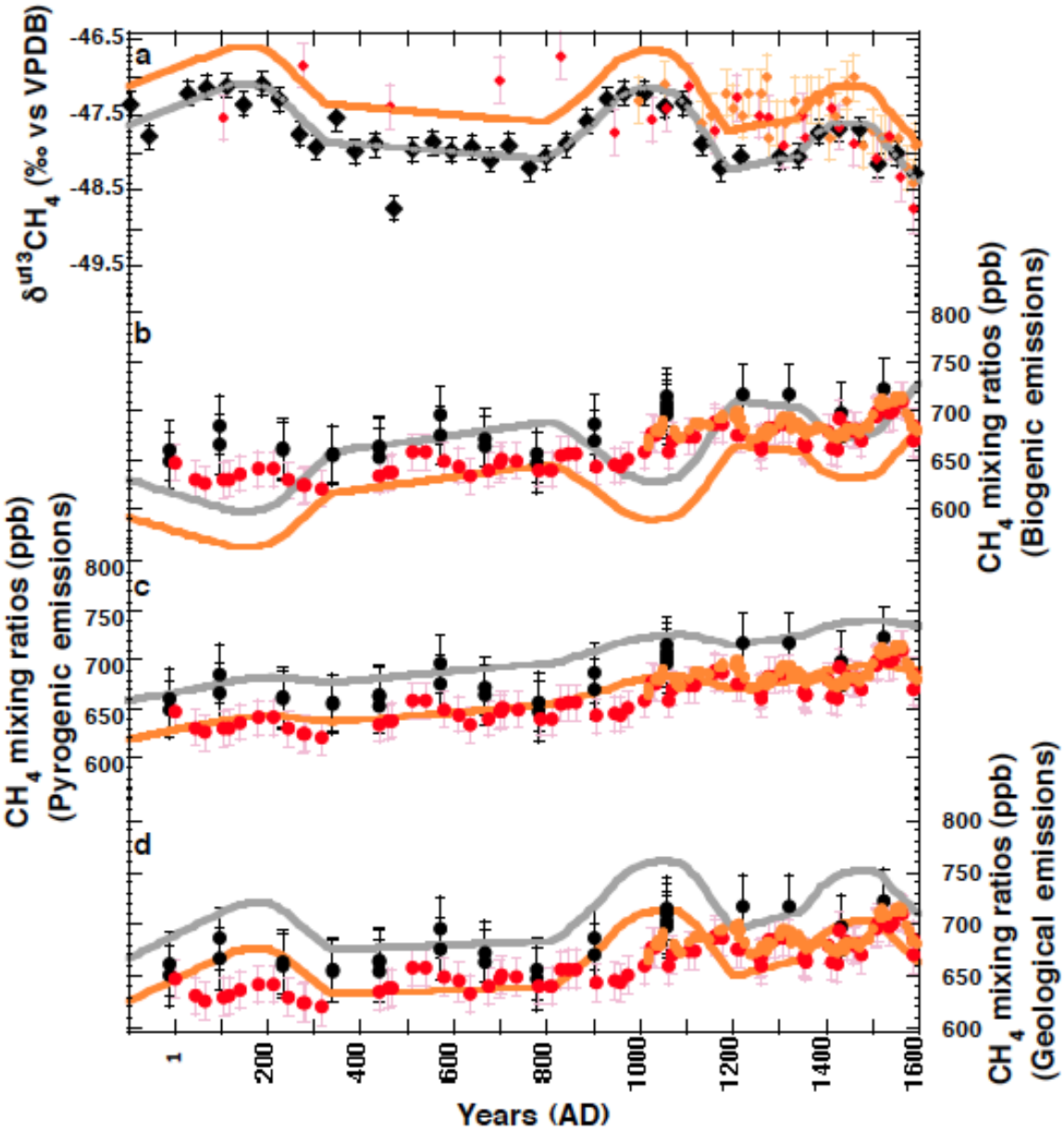
Supplementary Figure S1. $\delta^{13}C$ and CH_4 records from the NEEM and EUROCORE ice cores and two-box model "base scenarios". a) $\delta^{13}C$ data from NEEM (black diamonds), EUROCORE (blue diamonds) and published SH datasets from West Antarctic Ice Sheet Divide (WAIS)[9] (orange diamonds) and Law Dome[8] (red diamonds). b) CH_4 mixing ratios from GRIP[10,11](black circles), LAW DOME[12] (red circles) and WAIS[13] (orange circles). a) & b) the two-box model base scenario results

are depicted as grey lines for the NH and orange lines for the SH. Lower panels: Emission scenario for biogenic sources (c, green line), pyrogenic sources (d, yellow line), geological sources (e, brown line) and fossil fuel sources (e, purple line). Each data-point represents one measurement and the uncertainty bars represent one-sigma standard deviations, based on the reproducibility of the different measurement systems.



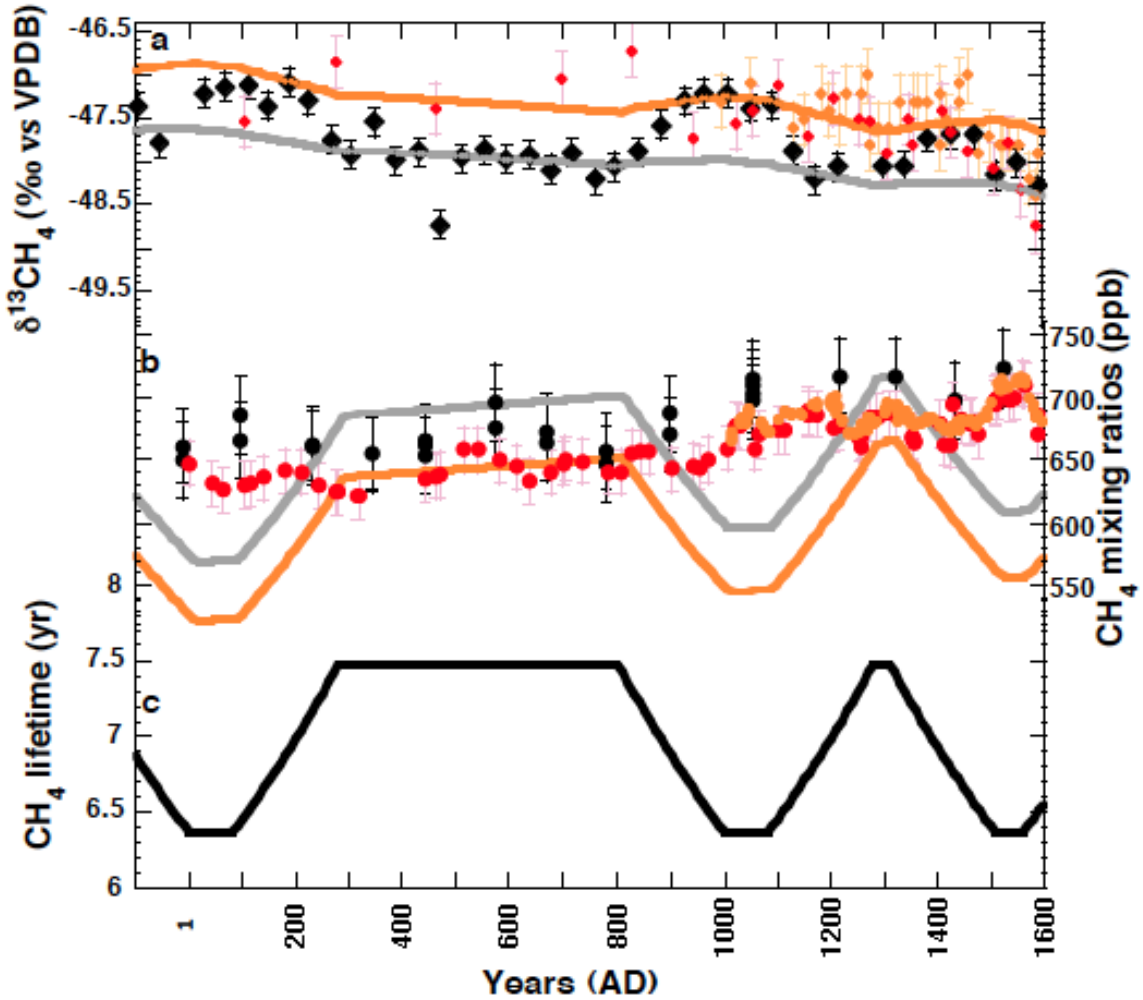
Suplementary Figure S2. Sensitivity of $\delta^{13}C$ and CH_4 mixing ratio to long-term changes in individual source categories. The strength of the individual sources and the lifetime are changed one by one in order to fit the observed long-term trend in CH_4 mixing ratios (a). Panel b-e show the effect of the changes of individual sources and the overall sink on the CH_4 isotopic composition, b: biogenic emissions. c: pyrogenic emissions. d: geological emissions. e: lifetime. Two-box model scenarios

are depicted as grey lines for the NH scenarios and as orange lines for the SH scenarios. Data: a: CH_4 mixing ratios from GRIP[10,11] (black circles) and Law Dome[12](red circles) and WAIS (orange circles)[13]. b-e: $\delta^{13}C$ from NEEM (black diamonds), EUROCORE (blue diamonds) and SH datasets: West Antarctic Ice Sheet Divide (WAIS)[9] (orange diamonds) and Law Dome[8] (red diamonds). Each data-point represents one measurement and the uncertainty bars represent one-sigma standard deviations, based on the reproducibility of the different measurement systems.

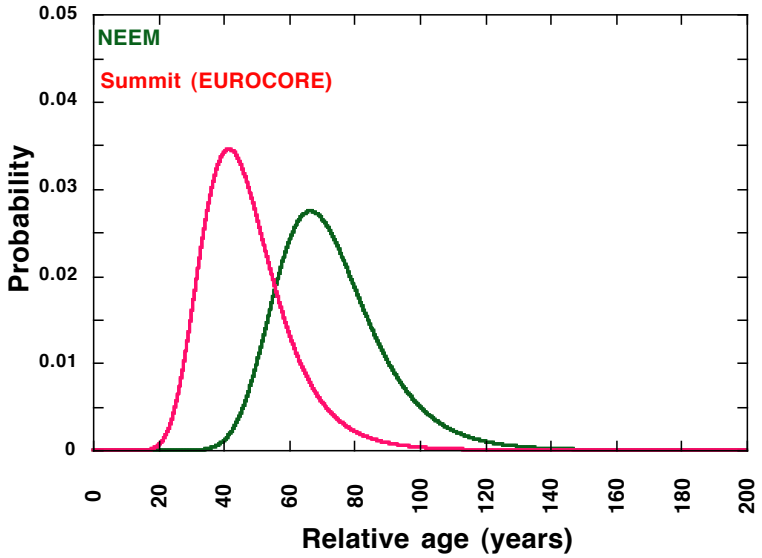


Supplementary Figure S3. Sensitivity of $\delta^{13}C$ and CH_4 mixing ratio to centennial-scale changes in individual source categories. The strengths of the individual source categories are changed one by one in order to fit the observed centennial-scale variability in $\delta^{13}C$ (a). The required changes in the CH_4 mixing ratio are compared to the observations (bottom panels) for changing biogenic emissions (b), pyrogenic emissions (c) and geological emissions (d). Two-box model scenarios are depicted as grey lines for the NH scenarios and as orange lines for the SH scenarios, which are

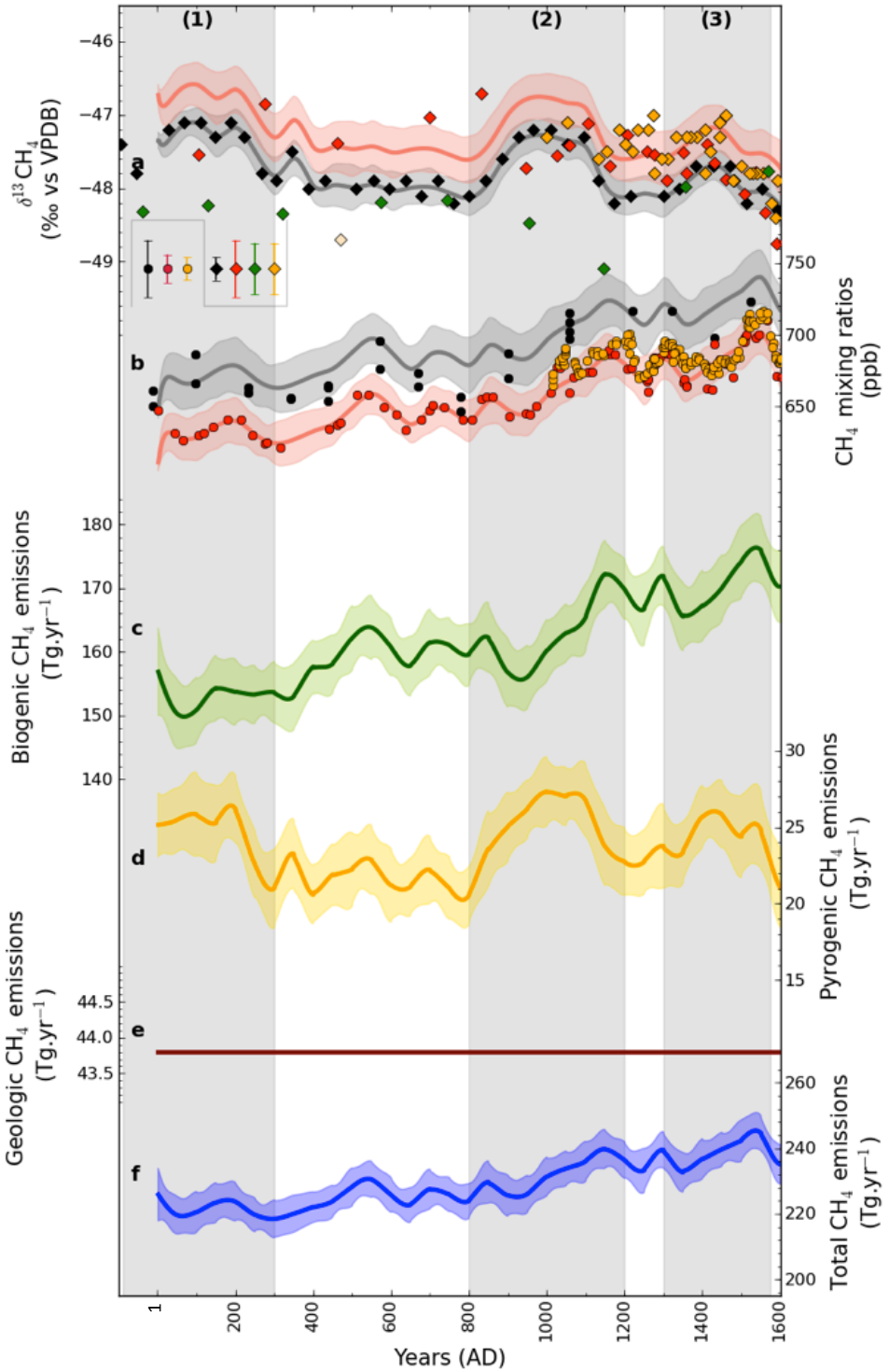
overlaid by the respective data. Data: a: $\delta^{13}C$ from NEEM (black diamonds), EUROCORE (blue diamonds) and SH datasets: West Antarctic Ice Sheet Divide (WAIS)[9] (orange diamonds) and Law Dome[8] (red diamonds). b-d): CH_4 mixing ratios from GRIP[10,11] (black circles), LAW DOME[12] (red circles) and WAIS[13] (orange circles). Each data-point represents one measurement and the uncertainty bars represent one-sigma standard deviations, based on the reproducibility of the different measurement systems.



Supplementary Figure S4. Sensitivity of $\delta^{13}C$ and CH_4 mixing ratio to centennial-scale variations in CH_4 lifetime. Two-box model scenarios are depicted as grey lines for the NH scenarios and orange lines for the SH scenarios. a) $\delta^{13}C$ data from NEEM (black diamonds), EUROCORE (blue diamonds) and SH datasets: West Antarctic Ice Sheet Divide (WAIS)[9] (orange diamonds) and Law Dome[8] (red diamonds). b) CH_4 mixing ratios from GRIP[10,11] (black circles), LAW DOME[12] (red circles) and WAIS[13] (orange circles). c) Underlying changes in CH_4 lifetime applied in the model (black line). Each data-point represents one measurement and the uncertainty bars represent one-sigma standard deviations, based on the reproducibility of the different measurement systems.

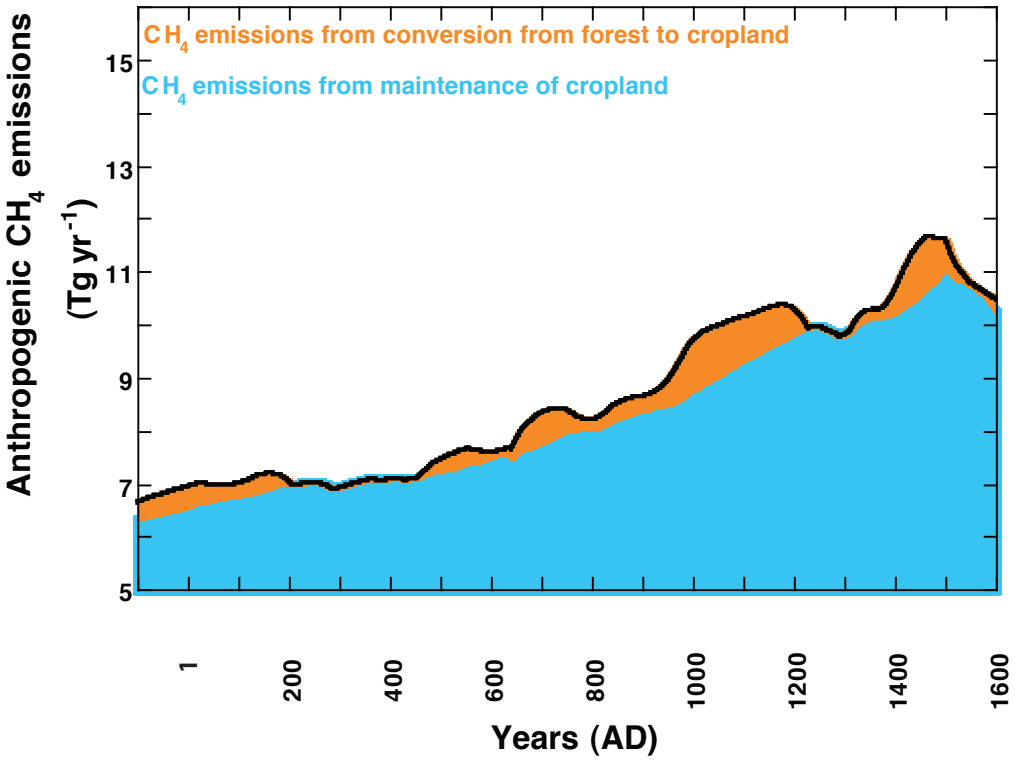


Supplementary Figure S5. Gas age distribution of air bubbles trapped in ice from NEEM (green) and EUROCORE (red). The x-axis represents the age of the air at the close-off depth where the air is totally isolated from the surrounding atmosphere.



Supplementary Figure S6. a) $\delta^{13}C$ measurements on air trapped in ice core from NEEM (black diamonds) (this study), EUROCORE (blue diamonds) (this study), GISPII[21] (green diamonds), LAW DOME[2] (red diamonds) and WAIS divide[3] (orange diamonds). b) CH_4 mixing ratios records from polar ice core from GRIP[6,7] (black), Law

Dome[2,5] (red) and WAIS[8] (orange). c) Reconstructed biogenic CH_4 emissions. d) Reconstructed pyrogenic CH_4 emissions. e) Fixed geological CH_4 emissions. f) Reconstructed total CH_4 emissions. Each data-point represents one measurement and the uncertainty bars represent one-sigma standard deviations, based on the reproducibility of the different measurement systems. The shaded red and grey areas in a) and b) represent the error ranges assigned to the CH_4 and $\delta^{13}C$ data, which are investigated to derive the uncertainties in the source reconstructions shown as shaded areas in c), d) and f). The uncertainty is described in section 1c.



Supplementary Figure S7. Calculated anthropogenic pyrogenic CH_4 emission scenario from the maintenance of cropland (blue) and from the conversion of forest to cropland (orange). The black curve shows the total anthropogenic pyrogenic scenario.